

Lab Objective:

- MATLAB and array operations.

Lab Preparation:

- Read slides and chapters 2 and 3 (first half) of the textbook.

Lab Procedures:

1, 3, 5, 7, 9, 11, 13, 15

With MATLAB, finish the odd number problems from 1 to 15, 21, and 23 in your textbook chapter 2 (section 2.11) and problems 1 to 3 of chapter 3 (section 3.9).

Lab Demonstration Requirements (40%):

Demonstrate at least 6 questions of procedures above in the lab computer during the lab time.

You can save all the code in one file and execute them step by step, Use pause command to generate a short pause between the questions.

After Lab Practice Questions (20%):

1. **Tracing Exercises:** Evaluate what will be the values of **p**, **q**, and **r** after these MATLAB instructions will be executed (**Do these by hand: Do not use MATLAB!**). Must show all your calculation procedures.

```
n1 = [1:5:9];  
n2 = [10 14 4];  
n3 = [length(n1) 0 0];  
p = [n1*3, n2, n3-n2];  
  
n4 = [10, 16, 3.3, 6];  
m1 = [1 2 3 4; 5 6 7 8];  
m2 = [m1;n4];  
q = ones (4,3) + m2';  
  
n5 = [10:20:60];  
m3 = [eye(3) .* 10; n5];  
s = [m3(3,2:3), m3(1:2,3)'];  
s(3) = 12;  
r = [m3(3,:), s];
```

Lab Report Submission Requirement (40%):

Show the demonstrations as required on the lab computers and submit Lab report based on the provided templates on the Blackboard. Please post all the screenshots of the inputs and running results of each step on the lab manual.